Course 1 · Week 4 — Two-group comparisons, associations, reporting

Cheatsheet — biostats_courses

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Choosing a comparison

Design	Continuous outcome	Binary outcome
Two independent groups	`t.test(y ~ g)` (Welch); Wilcoxon rank-sum	`prop.test` / `fisher.test`
Paired / pre-post	`t.test(pre, post, paired = TRUE)`; Wilcoxon signed-rank	McNemar
> 2 groups, continuous	`aov(y ~ g)`; Kruskal-Wallis	chi-square

Effect sizes

Statistic	What it reports
Mean difference + 95% CI	primary for continuous
Cohen's d	standardised mean difference
Hedges' g	small-sample correction of d
Risk ratio (RR) / odds ratio (OR)	two proportions
Risk difference	absolute, clinically intuitive

```
effectsize::cohens_d(y ~ g)
```

Two proportions

```
prop.test(c(tA, tB), c(nA, nB))          # asymptotic
fisher.test(matrix(c(tA, nA - tA,
                     tB, nB - tB), 2))   # exact, small cells
```

Report RR (or OR) with 95% CI, not just the p-value.

Correlation

Method	Captures	Assumptions
Pearson	linear association, continuous	bivariate normal, no outliers
Spearman	monotonic association	rank-based, robust
Kendall	concordant/discordant pairs	robust, slow on large data

```
cor.test(x, y, method = "spearman")
```

Non-parametric tests

Test	Replaces
Wilcoxon rank-sum (Mann-Whitney)	two-sample t
Wilcoxon signed-rank	paired t
Kruskal-Wallis	one-way ANOVA
Sign test	paired t, when even ranks fail

Power and sample size

```
pwr::pwr.t.test(d = 0.5, power = 0.80, sig.level = 0.05,
                type = "two.sample")
```

Family	Function
two-sample t	`pwr.t.test`
two proportions	`pwr.2p.test`, `pwr.2p2n.test`
correlation	`pwr.r.test`
one-way ANOVA	`pwr.anova.test`

Simulation-based power for anything the textbooks skip: `simr::powerSim`.

Reporting with `gtsummary`

```
library(gtsummary)
trial |>
  tbl_summary(by = arm, statistic = list(all_continuous() ~ "{mean} ({sd})")) |>
  add_p() |>
  add_overall()
```

Decision rule for Week 4

- Ask: one-group, two-group, paired, or many-group?
- Check: normal-ish, or should I use a rank-based test?
- Report: point estimate + 95% CI + effect size; p-value last, not first.

Common pitfalls

- Equal-variance t-test (default in some languages) when variances differ.
- Reporting OR when the audience expects RR (or vice versa).
- Forgetting that the p-value of a paired test depends on the pairing.
- Chaining correlations until one is "significant".

Further reading

- Altman, Machin et al., *Statistics with Confidence.*
- `gtsummary` docs.